

HLDNet: Abandoned Object Detection Using Hand Luggage Detection Network

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Abstract—Abandoned object detection has recently been studied to detect dangerous situations such as illegally dumping explosives. Existing abandoned object detection used background subtraction, which is sensitive to noise and clutters of a similar size in outdoor environments. Unlike existing approaches, this article presents a deep learning-based detection method that is suitable for use in outdoor environments because it is robust to ghost effects and illumination changes. The proposed method consists of three parts: 1) pedestrian detection and tracking; 2) hand luggage detection (HLD); and 3) abandoned object decision. Since the proposed method detects the abandoned object through HLD instead of directly detecting the abandoned object itself, we could successfully improve the detection accuracy and robustness. The proposed method can also detect dangerous weapons or restricted hand luggage through HLD. This article presents an intelligent abandoned object detection system that can be applied for consumer applications.

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■ **VARIOUS COMPUTER VISION** techniques have been proposed to analyze abnormal behaviors from video acquired by closed circuit television (CCTV) or internet protocol cameras. Abnormal behavior analysis is a key technology to build an intelligent surveillance system in smart city or

Table 1. Various Studies on Abandoned Object Detection Based on Background Subtraction.

Method	Reference
Background subtraction-based	3–5
Robust background subtraction-based	6–8
Background subtraction + Deep learning-based	8–10

smart village.^{1,2} Among various abnormal behaviors, abandoned object detection refers to decide if a pedestrian intentionally dumps an object at a certain place. Accurate analysis of abandonment behavior is required to determine crucial situations such as illegal garbage dumping or installing explosives in a public place.

Abandonment behavior is determined based on detection of an abandoned object. In existing abandoned object detection methods, an object is determined as an abandoned one if it stays at a fixed position for a prespecified time duration. For that reason, most abandoned object detection algorithms use a background subtraction method to detect the static object. Among them, the dual background subtraction is an improved method that uses two pieces of background information with different updating periods. It can better detect static objects than a simple background subtraction method since slowly changing foreground information is absorbed in the background. However, the dual method is still vulnerable to various challenging factors affecting the background, such as illumination change or camera shaking. Therefore, the detection method depending on the background information is not suitable for a pan-tilt-zoom (PTZ) camera or dynamic outdoor environment.

Most of the abandoned object detection methods used the background subtraction method, which removes a static background by detecting the difference between two adjacent video frames and using background modeling. In the field of abandoned object detection, it is known as a simple, efficient method that can extract moving objects from fixed background. Table 1 refers to studies of abandoned object detection based on various background subtraction. Unlike other abnormal behaviors such as violence or slip, there were few deep learning

approaches to the abandoned object detection problem. Although violence and slipping behaviors can be detected by itself, detection of abandonment requires a series of processes to train a dumping action and to determine whether the object is intentionally abandoned in a sufficient number of image frames. In this context, a deep learning approach can also classify the abandoned object.

We define that an abandoned object is dumped by a human hand. We added the condition since it is difficult to recognize the dumping situation using a simple background subtraction method. The background subtraction-based object detection methods has a fundamental problem of false detection due to dynamic background such as flickering neon signs and static objects without motion. To prevent such false detection, we define an abandoned object as an object dumped by a human. In this context, to detect an abandoned object after recognizing the dumping situation, the abandoned object should be detected by tracking the object in the hand. Although it is difficult to detect an unspecified abandoned object using a learning-based method, but it is possible to detect an object carried by a human hand. Therefore, we propose first locate the object located carried by the hand. Next, we classify it as a hand luggage class. We then determine abandonment behavior by detecting the hand luggage. More specifically, the unspecified hand luggage is detected using a deep neural network that merges keypoint and objects detection. To track the hand luggage, we propose a learning-based detection method that is robust to challenging factors, such as illumination changes and camera shaking.

The article is organized as follows. Section “Proposed Method” presents the proposed abandonment behavior detection algorithm based on the HLD. Section “Experimental Results” summarizes experimental results, and Section “Conclusion” concludes the article.

Proposed Method

This section describes the abandoned object detection system using learning-based hand luggage localization. The proposed system consists of three parts as shown in Figure 1: i) pedestrian



Figure 1. The flowchart of proposed method.

detection and tracking, ii) hand luggage localization, and iii) abandoned object decision. The proposed system was designed for a CCTV-based surveillance environment, and the detail of each part of the system will be given in three sections, respectively.

Pedestrian Detection and Tracking

The first part of the proposed system detects and tracks a pedestrian who potentially carries hand luggage to recognize the context of abandoned situation. Results of pedestrian detection and tracking can provide the distance between the hand luggage and the owner. We used the RetinaNet detector¹¹ to detect pedestrians, and a kernelized correlation filter (KCF)-based tracker^{12,13} to track them. In outdoor environments, many challenging factors cause tracking failure, such as changes in illumination, rapid movement, and occlusion when tracking pedestrians. The failure of pedestrian tracking in the proposed abandoned object detection system causes false detection of hand luggage, which results in failure in detecting abandoned objects.

As shown in Figure 2, the pedestrian detection and tracking module first estimates the coordinate of a pedestrian, and tracks it using a KCF tracker. The KCF tracker has the two advantages: i) efficient computation by replacing the correlation operation in the spatial domain with the element-by-element operation in the Fourier domain, and ii) robustness to shape changes using online training. In the surveillance system, it is necessary to check if a new pedestrian appears or a pedestrian being tracked does not exist in the image along with the tracking failure. RetinaNet detector was used to detect pedestrians with additional datasets acquired by the on-site CCTV environment to perform training. Because the tracker is updated through detection at a regular interval, retracking is possible by updating the detected pedestrian even if the tracking fails. In the proposed tracking method, the same pedestrian is retracked by comparing the similarity between the pedestrian being

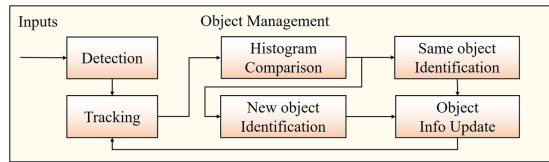


Figure 2. Block diagram of the pedestrian detection and tracking module, which is represented by the first block in Figure 1.

tracked and the detected pedestrian. To compare the similarity, we used a histogram-based method for robust tracking against challenging environment such as occlusion.¹⁴

HLD Network

Existing methods detected a static abandoned object using background subtraction, which results in false detection such as blinking signs and moving objects. To solve that problem, the proposed method tries to detect an abandoned object using a deep neural network that localizes hand luggage abandoned by a person. The proposed network merges two networks to detect an unspecified object located in the hand. The merged network consists of a keypoint detection network to detect the hand location and an object detection network¹⁵ to detect various objects including hand luggage. The keypoint detection process obtains a human skeleton by detecting the points of each joint. The obtained skeleton plays a role in recognizing an action. In this article, we used the OpenPose network¹⁶ for the latter purpose. The OpenPose network computes a confidence map to locate joints and an affinity field to connect the joints. The human skeleton can be obtained by connecting the confidence map of the joint along the direction of the affinity field. This algorithm improves runtime performance and accuracy in the sense of part affinity fields (PAF). In this article, this keypoint detection and SSD object detection are merged to detect hand luggage. The SSD network is widely used in a surveillance system since it can detect an object detection in real time.

In the object detection network, the feature map is divided into multiple scales to detect objects. There are six feature maps of sizes 38×38 , 19×19 , 10×10 , 5×5 , 3×3 , and 1×1 . The reason for dividing the scale is to detect

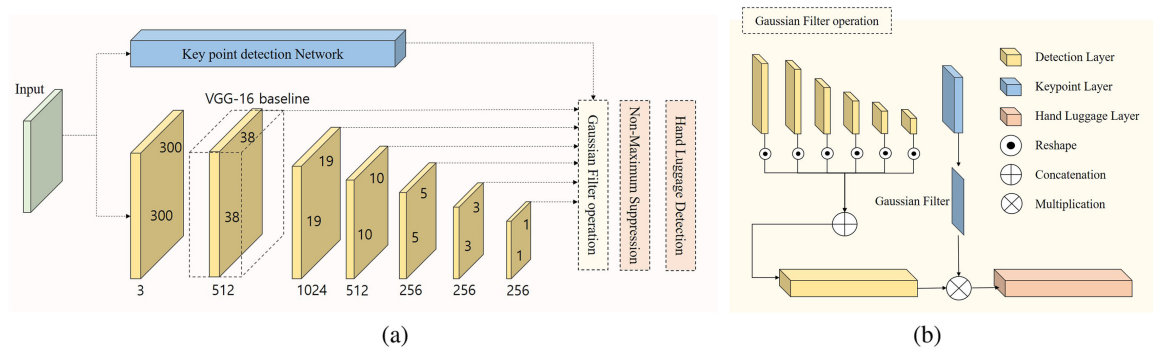


Figure 3. The proposed HLD network.

objects of different sizes for each scale. Objects detected through the six feature maps have location information and confidence values. In the keypoint detection network, VGG-19 was used to detect hand location information, and it is used in the multibox layer shown in Figure 3 to detect the hand luggage along with the object information detected in the object detection network.

Figure 3 shows the structure of the proposed HLD network. The proposed network structure is shown in Figure 3(a), which combines the keypoint detection and object detection networks. The keypoint detection network uses a pre-trained OpenPose network¹⁶ to detect the location of the hand. In the existing SSD-based approach,¹⁵ nonmaximum suppression is performed after concatenation in the multibox layer. On the other hand, the proposed method performs filtering to detect hand luggage in the Gaussian filter operation layer. Figure 3(b) shows the structure of the proposed Gaussian filter operation, where feature maps with different scales passing through the existing SSD object detection layer are concatenated into one object detection layer. Layers divided into six scales are concatenated into one layer before NMS is performed. The concatenated layer has 8732 coordinate information and confidence information, which are then converted into coordinate values suitable for each scale after the NMS process. After passing through the key point detection network, a Gaussian filter is generated based on the location of the hand. The weighted detection tensor is generated by calculating the generated Gaussian filter and object detection tensor. These two tensors are used to detect hand luggage through the operation with a

Gaussian filter.¹⁷ The generated Gaussian filter is used to adjust the confidence value of the detected object that does not exist at the location of the hand. In this work, the Gaussian-weighted confidence value is defined as

$$C_G = G(x, y)C(x, y) \quad (1)$$

where $G(x, y)$ is the Gaussian filter and $C(x, y)$ is the confidence value of the detected object. To compensate for the case where the luggage covers the hand or the hand is not detected, a Gaussian filter is applied to the elbow region. Due to the Gaussian filter generated at the hand location, the closer the hand location, the more the value of confidence is preserved. Otherwise, it converges to zero. In this context, objects other than hand luggage are not detected or gives a low confidence values. In the proposed method, it is ranked according to the object's confidence to detect candidate hand luggage objects. Figure 4 shows distributions of both confidence values and its Gaussian filtered version. Figure 4(a) shows the original input image, where a person carries a hand luggage. Figure 4 (b) graphically shows the Gaussian filter whose center is aligned to the location of the person's right hand. Figure 4(c) shows the result of HLD using the existing SSD method, and generates many erroneously detected regions because of inaccurate, spurious confidence values shown in Figure 4(d). Figure 4(e) shows the result of a successful detection of the luggage using Gaussian filtered confidence values given in Figure 4(f).

Figure 5 shows the results of SSD and the proposed HLDNet trained with a single hand luggage class. In the experimental results, the detection

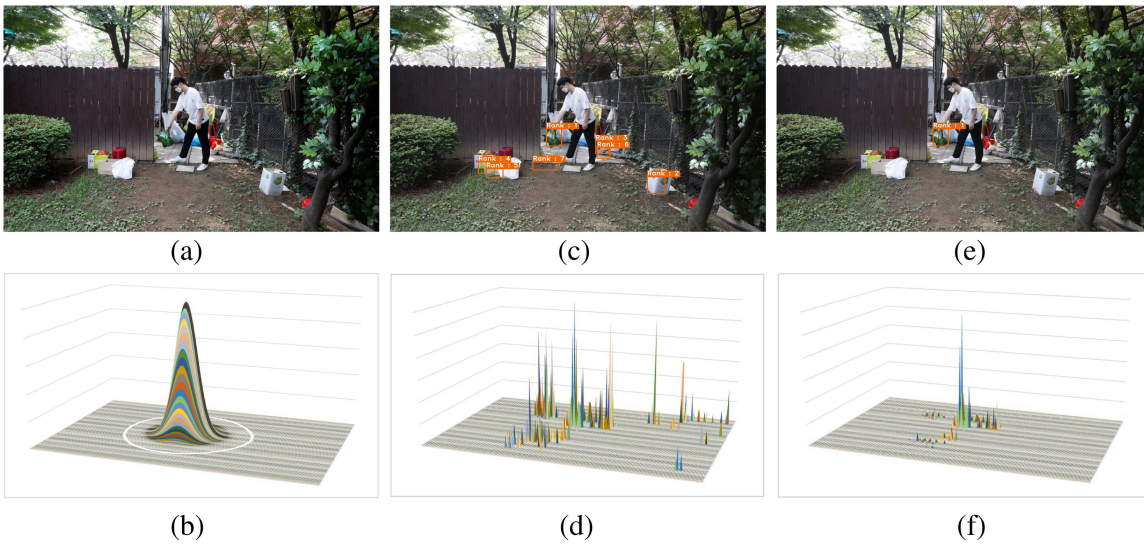


Figure 4. HLD results using Gaussian filtered confidence values. (a) Input image. (b) Gaussian filter. (c) The result of object detection using the existing SSD method. (d) Confidence values without applying the Gaussian filter given in (b). (e) The result of luggage detection using the proposed method. (f) Confidence values filtered by the Gaussian filter given in (b).

results were calculated by ranking them in the order of the highest confidence values. There are two approaches to detect hand luggage using only SSD: i) detecting the object nearest to the hand and ii) detecting the object with the highest confidence value. The first approach may fail with objects around the hand. The second approach also fails with other objects having higher confidence than hand luggage as shown in Figure 5(a). To solve this problem, we propose a novel Hand Luggage Detection Network (HLDNet) using a Gaussian filter that can reasonably combine the two approaches. As shown in Figure 5(b), HLDNet can successfully detect only hand luggage. Figure 5(b) shows the accurate detection of the hand luggage using the proposed algorithm. The proposed method detects only hand luggage and filters out other objects.

It can be embedded in object detection networks such as YOLO¹⁸ and faster-RCNN.¹⁹

To train the existing SSD network, we used 21 classes in datasets of COCO²⁰ trainval35 k and Pascal VOC 2007+2012. However, it is challenging to train an unspecified, abandoned object as a specific class. To overcome this problem, we trained various objects located in or right below the hand as a single class to detect unspecified objects. We used Korea Internet Security Agency (KISA)²¹ and Chung-Ang University (CAU) datasets. The abandoned object datasets, such as plastic bags, bags, and paper boxes, were constructed and trained using KISA and in-house CAU datasets. The proposed method detects various objects using the existing SSD network and then detects the hand luggage using the Gaussian filter generated using the OpenPose



Figure 5. The HLD results. (a) SSD detection method. (b) Proposed method.

network. The OpenPose network used a model that is pretrained with COCO datasets to detect hand location. The training of the proposed model follows the same strategy used in the existing SSD to detect various objects. The proposed network takes images as input to the OpenPose network and the SSD object detection network, respectively, as shown in Figure 4. In the SSD object detection network, training is performed by resizing to 300×300 size and configuring the vgg-16 layer and 5 extra layers. The training parameters include batch size 16, learning rate 10^{-3} , momentum 0.9, and weight decay 0.0005. We train the network for 120 000 iterations, which takes about 21 hours on an RTX 2080Ti GPU. The training of the proposed model follows the same strategy used in the SSD.

The loss function in the SSD is a combination of i) the cross entropy loss, denoted as L_{conf} and ii) bounding box regression loss, denoted as L_{loc}

$$L(x, c, l, g) = \frac{1}{N} (L_{\text{conf}}(x, c) + \alpha L_{\text{loc}}(x, l, g)). \quad (2)$$

In (2), L_{conf} is given as

$$L_{\text{conf}} = - \sum_{i \in \text{Pos}} x_{ij}^p \log(\hat{c}_i^p) - \sum_{i \in \text{Neg}} \log(\hat{c}_i^0) \quad (3)$$

where

$$\hat{c}_i^p = \frac{\exp(\hat{c}_i^p)}{\sum_{\text{Pos}} \exp(\hat{c}_i^p)} \quad (4)$$

and x_{ij}^p means that the i th default box in a certain grid matches the j th ground truth box in the p th class, and c represents the confidence of multiple classes.

In (2), L_{loc} is given as

$$L_{\text{loc}}(x, l, g) = \sum_{i \in \text{Pos}} \sum_{m \in \{cx, cy, w, h\}} x_{ij}^k S_{L1}(\ell_i^m - \hat{g}_j^m) \quad (5)$$

where

$$\hat{g}_j^{cx} = (g_j^{cx} - d_i^{cx}) / d_i^w \quad (6)$$

$$\hat{g}_j^{cy} = (g_j^{cy} - d_i^{cy}) / d_i^h \quad (7)$$

$$\hat{g}_j^w = \log\left(\frac{g_j^w}{d_i^w}\right), \hat{g}_j^h = \log\left(\frac{g_j^h}{d_i^h}\right) \quad (8)$$

where S_{L1} is the robust bounding box regression loss using ℓ_1 smoothing, which is known to be efficient to predict a rectangular region using

regression. l represents the predicted box, (cx, cy) the center coordinate, d the default bounding box, $\{w, h\}$ the width and height of the bounding box, $\hat{g}_j^{cx}$ and $\hat{g}_j^{cy}$ are normalized boxes with respect to cx , and cy , $\hat{g}_j^w$, and $\hat{g}_j^h$ are normalized boxes with respect to the width and height of the default box. We modified the pretrained VGG-16 as a feature extractor. Six extra layers minimize two loss functions: a confidence loss using softmax in (2) and an $L1$ loss that calculates the offset of the bounding box. In SSD, a default anchor box in Faster R-CNN was used

$$s_k = s_{\min} + \frac{s_{\max} - s_{\min}}{m - 1} (k - 1), k \in [1, m] \quad (9)$$

$$a_r \in \{1, 2, 3, \frac{1}{2}, \frac{1}{3}\} \quad (10)$$

$$w_k^a = s_k \sqrt{a_r}, h_k^a = s_k / \sqrt{a_r} \quad (11)$$

where S represents the scale of the box, k an integer from 1 to 6, and a_r a prespecified aspect ratio. After determining the offset of the default box using (9), the IOU is compared with each grid cell of the feature map and the GT, and the learning proceeds by selecting those with a threshold value of 0.5 or more. We selected only bounding boxes with high confidence loss to maintain the ratio of positive and negative as 1:3. Based on the reliability between the selected default box and the abandoned object class, the confidence loss of (3) is obtained. Bounding box regression is performed using the final confidence score and bounding box coordinates. After learning, the objects with overlapping offsets are removed through the NMS operation, and the reliability of detected objects far from the hand position is adjusted through Gaussian filtering to detect the final hand luggage. Figure 6 shows the training curve of the proposed network.

Abandoned Object Detection System

In this section, we present an abandoned object decision system using the previously detected hand luggage region in section “HLD Network.” Figure 7 shows the framework of the proposed abandoned object decision system. SSD can detect only luggage in hand, but it cannot detect an abandoned object. The proposed method combined the SSD detector with KCF

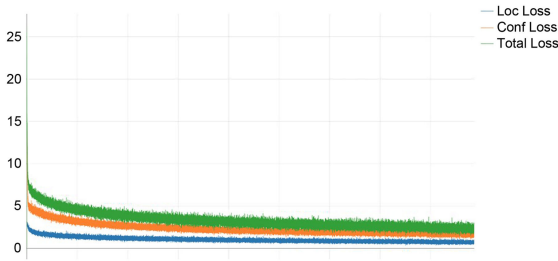


Figure 6. Proposed Network Training Curve.

tracker to detect the moment of abandoning by calculating the distance between the hand and luggage. Specifically, we need to trace the coordinate of the hand luggage using the high-speed KCF tracker. However, there are various factors of false detection of hand luggage, such as occlusion, noise, and image shaking while tracking pedestrians. In the proposed system, hand luggage regions are sorted to remove falsely detected regions. Falsely detected objects from the pedestrian are usually caused by occlusion, and unlike hand luggage, they are not detected in continuous frames. On the other hand, an action of pedestrians dumping hand luggage is shown in a set of consecutive frames, which will be shown in detail in section “Experimental Results.” In the proposed system, the detected hand luggage is not immediately tracked, but stored for comparison with the next frame. The hand luggage detected in the next frame uses the histogram comparison with the previous frame, through which the similarity between the two objects is measured. The histogram comparison used as a similarity comparison method compares color values in the hue channel. This method is relatively simple and efficient with less computational complexity than template matching or feature matching methods. If the similarity is higher than the specified threshold using this the histogram comparison, the object

is determined as a candidate of the abandoned luggage counted. In this system, if a similar object is detected more than five times in a row, it is determined as hand luggage to judge abandonment, and this system is also used for retracking when hand luggage tracking fails. We measure the distance between the abandoned luggage and the owner using the sorted hand luggage using the KCF tracker¹² to measure the distance from the owner. The KCF tracker used as a tracking algorithm is a method that does not use deep learning and is a suitable for algorithm as a tracking method in this article, because it shows fast speed and outstanding acceptable accuracy.

Finally, to determine the abandoned object, the distance between the owner and the hand luggage being tracked is calculated as shown in Figure 8. We cited Luna’s survey paper²² for the conditions, and verified them through experiments, where W is the owner’s width and D is the distance from the hand luggage. In the proposed system, if the distance (D) is more than two times larger than the width (W) and this state is maintained for 5 s, it is determined as an abandoned object.

Experimental Results

In this article, the abandoned object is detected based on the detected hand luggage using the network proposed in section “Proposed Method.” This section describes the experimental results using the ABODA datasets,²³ and compares the performance of various methods in the sense of intersection over union (IOU), which is a good indicator to demonstrate the performance of the HLDNet.

The ABODA datasets are often used to evaluate the performance of abandoned object detection. These datasets were created by Lin, and scenarios in the datasets include outdoor, night, illumination change, indoor, and crowded scenes.²⁴ Table 2

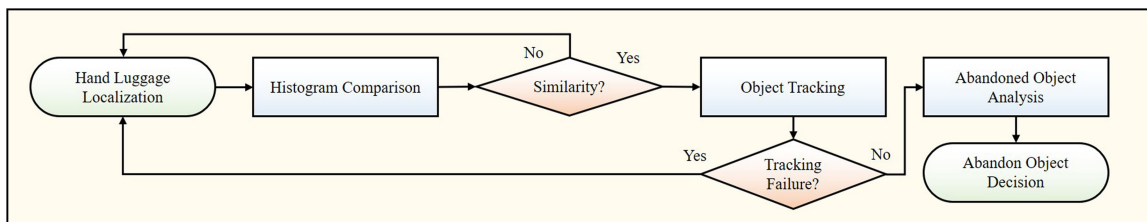


Figure 7. Abandoned Object Decision System Framework.



Figure 8. Abandoned Object Decision.

shows the comparative experimental results between the proposed HLDNet and existing abandoned object detection methods. The proposed method successfully detected abandoned objects in ten test videos. It shows the best result like Park's and Shyam's methods. Park's method detected static objects using background modeling and composed a system to complement the problem vulnerable to illumination change to detect the abandoned object. Shyam's method also detected static objects using background modeling, but unlike Park's method, the learning-based detection method was additionally used to detect the abandoned object. Shyam's detection network use backpacks and handbags as a training dataset to detect candidate objects. Learning-based methods can detect a specific object, but is not suitable for detecting an unspecified abandoned object. In this context, Shyam's method shows superior results in the ABODA dataset because most abandoned objects in the dataset exist as a specific object such as a handbag or a backpack. However, in a real environment where abandoned objects exist as various objects, the performance of detecting abandoned objects is not promising.

In existing background subtraction approaches, it is difficult to accurately detect an abandoned or small objects due to a crowd. Because the abandoned object size is small, the background subtraction-based method may classify the abandoned object as noise, and it is also vulnerable to the object occlusion problem caused by crowds. The proposed network, like other research, also failed in detecting abandoned objects in video 11 because of an inaccurate keypoint detection in small objects and the occlusion problem. The proposed system does not use background subtraction, but a deep learning-based method. The performance of abandoned object detection is significantly limited in a crowded situation with more than ten people, which is considered as a chronic problem in the field of computer vision. However, the proposed method still works since the action of illegally abandoning object usually occurs in a secluded place. This problem can be solved either by improving the detection performance or restoring the quality of input video. However, there are still detection failures as shown in Figure 9 and Table 3. As shown in Table 2, recent abandoned object detection methods use a dual background modeling-based approach. The dual background modeling method is efficient for detecting static objects, but frequently fails because of various challenging issues. Figure 9 shows that the dual background modeling method using KISA datasets is difficult to apply in a real outdoor environment. KISA datasets includes many outdoor scenarios with dynamic background such as wave in beach, moving

Table 2. Performance of Abandoned Luggage Detection Using Various Methods on ABODA Dataset.

video	Senario	GT	HLDNet		Park [7]		Lin [24]		Wahyono [25]		Patrick [26]		Wentong [27]		Shyam [10]	
			TP	FP	TP	FP	TP	FP	TP	FP	TP	FP	TP	FP	TP	FP
V1	Outdoor	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0
V2	Outdoor	1	1	0	1	0	1	0	0	1	1	0	1	0	1	0
V3	Outdoor	1	1	0	1	0	1	0	0	1	1	0	1	0	1	0
V4	Outdoor	1	1	0	1	0	1	0	0	1	1	0	1	0	1	0
V5	Night	1	1	0	1	0	1	1	1	0	1	0	1	0	1	0
V6	Illumination	2	2	0	2	0	2	0	-	-	1	0	2	1	2	0
V7	Illumination	1	1	0	1	0	1	1	-	-	1	1	1	0	1	0
V8	Illumination	1	1	0	1	0	1	1	-	-	1	0	1	1	1	0
V9	Indoor	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0
V10	Indoor	1	1	0	1	0	1	0	1	0	1	0	1	1	1	0
V11	Crowded	1	-	-	-	-	1	3	-	-	1	1	1	1	-	-

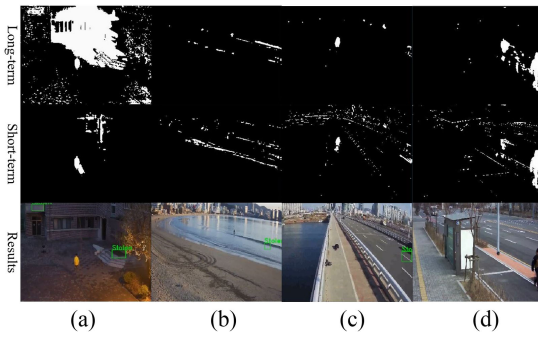


Figure 9. The vulnerable results of the dual background modeling method in the outdoor environment.

vehicles on the road, and low-light conditions.²¹ Figure 9(a) shows experimental results in low light environments. In short-term and long-term images, it can confirm that the background information varies greatly even with small changes in illumination. Since the dual background modeling is very sensitive to changes in illumination, abandoned object could not be correctly detected. Figure 9(b) shows experimental results in the beach with periodic waves. In the dual background modeling method, the background should be maintained as the stationary area with little change. However, in short-term and long-term images, the background region is not properly generated due to the periodic waves. For this reason, Figure 9(b) shows the false detection result. Figure 9(c) shows experimental results of images taken on the bridge. This result also shows that the background area was not correctly generated due to moving vehicles. In the dual background modeling method, there is a case where a stopped vehicle is considered as an abandoned object. Figure 9(c) shows the result of detecting a stopped vehicle as being classified as an abandoned object. Figure 9(d) shows the experimental result of the abandonment of a small object. In the dual background modeling method, noise results in erroneous detection. To solve this problem, noise is removed before detection. In case of small objects as shown in Figure 9(d), the small object cannot be detected. Figure 9 shows several problems caused by the dual background modeling method. In addition to the listed scenarios, there are many challenging factors that cause false

Table 3. Comparative Experimental Results Between the Proposed Method and the Background Modeling Method in Fig. 9.

Video	Scene	GT	Dual Background	HLDNet
(a)	Illumination	0	3	0
(b)	Wave	0	2	0
(c)	Moving Car	0	2	0
(d)	Small Object	1	0	1

detection by being influenced by background information such as weather, flickering neon sign, and stationary pedestrian. For this reason, background subtraction-based methods are not suitable for use in real environments due to the disadvantages of the dual background modeling method. HLDNet, which is a learning-based method, is more suitable than background subtraction-based methods in real environments because it can learn complicated situation using a large set of training data.

Table 3 summarizes the performance in the experiment of Figure 9. In Figure 9(a), (b), and (c), there is no abandoned object under a challenging condition. The dual background modeling method results in a false detection because of incorrect background information, which implies the inherent problem of the background modeling method. On the other hand, the proposed method can selectively detect only abandoned objects without false detection. Figure 9(d) shows the detection results of a small abandoned object. More specifically, the proposed method determines an abandoned object based on the detection of human hand luggage, which represents a high-level context information. Figure 9 and Table 3 show that there are many problems in using the background modeling method in the real environment, but the proposed learning-based abandoned object detection method gives better performance for real applications.

The proposed method proves its performance using hand luggage and abandoned object detection accuracies. The former is calculated based on the IOU, which is widely used to evaluate the performance of object detection. The latter is

Table 4. IOU Results of Proposed Method. (a) IOU Values When the Owner is Carrying Hand Luggage. (b) IOU Values for Five Frames in the Dumping Situation.

Video	Scenario	(a)	(b)
V1	Outdoor	0.6487	0.8186
V2	Outdoor	0.8344	0.8903
V3	Outdoor	0.8446	0.8788
V4	Outdoor	0.8838	0.9398
V5	Night	0.6838	0.8210
V6	Illumination change	0.6782	0.8315
V7	Illumination change	0.8479	0.8940
V8	Illumination change	0.6617	0.8251
V9	Indoor	0.8682	0.9372
V10	Indoor	0.8872	0.9529
V11	Crowded	–	–
Average		0.7839	0.8789

calculated based on the number of successes, which is widely used in existing abandoned object detection methods. Unlike conventional abandoned object detection methods, the proposed method determines abandoned objects based on the hand luggage. The accurately detected hand luggage provides a high success rate in detecting an abandoned object. In this context, the proposed method requires a high IOU value since it tracks the detected hand luggage. Table 4(a) shows the IOU values when the owner is carrying hand luggage in each scenario. In this experiment, the average of the IOU was equal to 0.7839. Because IOU result in (a) is walking after carrying hand luggage, it shows lower results than (b) due to various factors such as occlusion but, which is sufficient to track hand luggage. To detect an abandoned object, an accurate detection of dumping situation is the most important. The histogram of the detected hand luggage is compared to determine whether it is the same object for five times in a row, and then tracking is performed. Figure 10 shows the results of detection in five consecutive frames in a hand luggage dumping situation. It can confirm that the proposed method shows outstanding tracking results every frame. Table 4(b) shows IOU values for five frames in the hand luggage

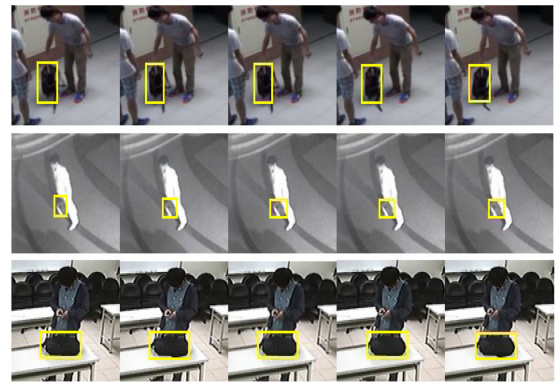


Figure 10. Results of an abandoned object in five frames.

dumping situation shown in Figure 10. The average IOU was 0.8789, which shows the outstanding results in the hand luggage dumping situation.

Conclusion

In this article, we proposed a system that detects an abandoned object based on the detected hand luggage using a deep neural network. The proposed system first tracks the pedestrian to detect the pedestrian's hand luggage. To overcome the challenging issues in detecting an unclassified hand luggage, we combined a keypoint detection and object detection networks. The abandoned object is finally detected using the distance between the detected hand luggage and owner. Since the proposed system can recognize the dumping situation based on the distance between the owner and the hand luggage, the proposed two-step method was verified from the experimental results that there were fewer false detection problems than existing single-step abandoned object detection methods. The performance of the proposed network was verified by measuring the IOU of the detected hand luggage, and the experimental results using ABODA dataset also showed better results than existing methods. Although the proposed method fails in a crowded environment scenario, it can be improved in the future research by improving the performance of pedestrian detection.

The proposed method is a learning-based detection method that does not use background subtraction and can overcome the disadvantages of using background subtraction-based detection

methods. It is suitable for outdoor environments or PTZ cameras. It can provide economical consumer security environment in both public and private areas in the form of a visual surveillance system. Since hand luggage may contain weapons or bombs, its detection and recognition is important to indicate dangerous situations. Also, because the proposed method detects abandoned objects based on the information of pedestrians, it can be used by integrating them into a system for detecting abnormal behaviors such as intrusion, loitering, and violence.

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